

ADVANCED QUANTUM MECHANICS

Exercise 13 - – presentation session – 27.01.2026



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1. Single band tight-binding model

We consider the tight-binding Hamiltonian with periodic boundary conditions $\hat{c}_{N,\sigma} \equiv \hat{c}_{0,\sigma}$

$$\hat{H}_0 = -t \sum_{j=0}^{N-1} \sum_{\sigma} \left(\hat{c}_{j+1,\sigma}^{\dagger} \hat{c}_{j,\sigma} + \hat{c}_{j,\sigma}^{\dagger} \hat{c}_{j+1,\sigma} \right) - \mu \sum_{j=0}^{N-1} \hat{n}_j.$$

It is convenient to Fourier transform the fermionic operators. The inverse transform is

$$\hat{c}_{j\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{+ikx_j} \hat{c}_{k\sigma}, \quad \hat{c}_{j\sigma}^{\dagger} = \frac{1}{\sqrt{N}} \sum_k e^{-ikx_j} \hat{c}_{k\sigma}^{\dagger}.$$

where $x_j = ja$, $\sigma = \uparrow, \downarrow$.

(i) Show that the allowed crystal momenta imposed by periodic boundary conditions are

$$k = \frac{2\pi}{Na} m, \quad m = 0, 1, \dots, N-1.$$

(ii) Show that:

$$\{\hat{c}_{k\sigma}, \hat{c}_{k'\sigma'}^{\dagger}\} = \delta_{kk'} \delta_{\sigma\sigma'},$$

and that all other anticommutators vanish.

(iii) Show that

$$-\mu \sum_j \hat{n}_j = -\mu \sum_{k,\sigma} \hat{c}_{k\sigma}^{\dagger} \hat{c}_{k\sigma}.$$

(iv) Show that

$$\hat{c}_{j+1,\sigma}^{\dagger} \hat{c}_{j\sigma} = \sum_k e^{-ika} \hat{c}_{k\sigma}^{\dagger} \hat{c}_{k\sigma},$$

and similarly,

$$\sum_j \hat{c}_{j\sigma}^{\dagger} \hat{c}_{j+1,\sigma} = \sum_k e^{ika} \hat{c}_{k\sigma}^{\dagger} \hat{c}_{k\sigma}.$$

(v) Collecting all terms show that the tight-binding Hamiltonian is diagonal in momentum space and spin:

$$\hat{H}_0 = \sum_{k,\sigma} [-2t \cos(ka) - \mu] \hat{c}_{k\sigma}^{\dagger} \hat{c}_{k\sigma} \equiv \sum_{k,\sigma} \varepsilon(k) \hat{c}_{k\sigma}^{\dagger} \hat{c}_{k\sigma},$$

with dispersion relation

$$\varepsilon(k) = -2t \cos(ka) - \mu.$$

Draw the band diagram within the first Brillouin zone.

2. Two band tight-binding model

Consider the tight-binding Hamiltonian for two bands, with s and p orbitals

$$\hat{H}_0 = - \sum_{j=0}^{N-1} \sum_{\sigma} \left[t_{ss} \hat{s}_{j\sigma}^{\dagger} \hat{s}_{j+1,\sigma} + t_{pp} \hat{p}_{j\sigma}^{\dagger} \hat{p}_{j+1,\sigma} + t_{sp} \hat{s}_{j\sigma}^{\dagger} \hat{p}_{j+1,\sigma} + t_{ps} \hat{p}_{j\sigma}^{\dagger} \hat{s}_{j+1,\sigma} \right].$$

Here

- t_{ss} arises from overlap of neighboring s orbitals,
- t_{pp} from overlap of neighboring p orbitals,
- $t_{sp} = t_{ps}^*$ describes s - p hybridization between adjacent sites.

(i) Use the inverse Fourier transforms to write the k -space Hamiltonian

$$\hat{s}_{j\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{+ikx_j} \hat{s}_{k\sigma}, \quad \hat{p}_{j\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{+ikx_j} \hat{p}_{k\sigma}.$$

(ii) Show that the Hamiltonian can be written as

$$\hat{H}_0 = \sum_{k,\sigma} \begin{pmatrix} \hat{s}_{k\sigma}^{\dagger} & \hat{p}_{k\sigma}^{\dagger} \end{pmatrix} \begin{pmatrix} \varepsilon_s(k) & V_{sp}(k) \\ V_{sp}^*(k) & \varepsilon_p(k) \end{pmatrix} \begin{pmatrix} \hat{s}_{k\sigma} \\ \hat{p}_{k\sigma} \end{pmatrix},$$

(iii) Derive the energies and couplings

$$\varepsilon_s(k) = \varepsilon_s - 2t_{ss} \cos(ka), \quad \varepsilon_p(k) = \varepsilon_p - 2t_{pp} \cos(ka),$$

and

$$V_{sp}(k) = -2t_{sp} \cos(ka)$$

3. Two-band tight binding: emergence of bandgap

Consider problem as before.

(i) Diagonalize the 2×2 Hamiltonian to find the two energy bands

$$E_{\pm}(k) = \frac{\varepsilon_s(k) + \varepsilon_p(k)}{2} \pm \sqrt{\left[\frac{\varepsilon_s(k) - \varepsilon_p(k)}{2} \right]^2 + |V_{sp}(k)|^2}.$$

(ii) Discuss the case where $|V_{sp}(k)|^2$ is very small,

(iii) Discuss the case where all $t_{ss} = t_{pp} = t_{sp} = t_{ps} = t$. Draw the band diagrams to scale for $t = 1$.